

11 (**short**) Two rigid discs with moment of inertia J are mounted on three light elastic shafts with torsional stiffness k as shown in Fig. 7. The outer ends of the system are clamped.

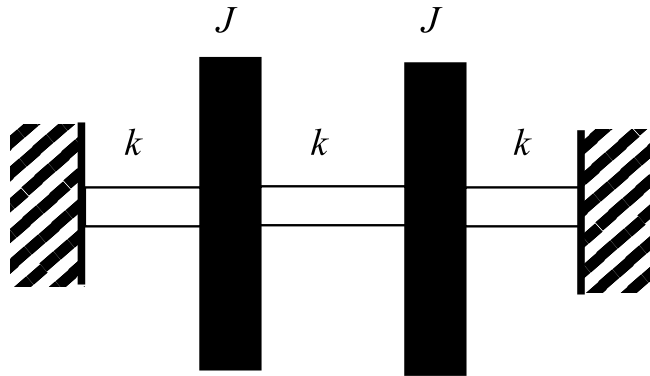


Fig. 7

- (a) Determine the natural frequencies and normal modes of this system by inspection. [5]
- (b) Describe the behaviour of the system for the initial conditions $\theta_1 = 1$ rad, $\theta_2 = 0$ rad, $\dot{\theta}_1 = 0$ rad s⁻¹, $\dot{\theta}_2 = 0$ rad s⁻¹, where θ_1 and θ_2 are the angular positions of the two discs. [5]